SERRE FORMULA: INTERSECTION PRODUCTS OF SHEAF CYCLES VIA ALTERNATING TOR SUMS

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Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file intersection.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate.

Editorial note. Occupation scan: combined claim_key + statement_anchor + generic-line scan over all registration files (active and withdrawn) found 0 hits for this chapter (is-occupied.json, 2026-10-14); consistent with the duty manager's six-check dispatch.

Source reference. Original reference 02QZ: sources/stacks-chow/tags.txt, source lines 3606–3606. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 00L0: sources/stacks-chow/tags.txt, source lines 770–770. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0B0T: sources/stacks-chow/tags.txt, source lines 13871–13871. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0B0Q: sources/stacks-chow/tags.txt, source lines 13868–13868. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0B03: sources/stacks-chow/tags.txt, source lines 13846–13846. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 00KQ: sources/stacks-chow/tags.txt, source lines 760–760. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let X be a nonsingular variety. Let $\mathcal{F}$ and $\mathcal{G}$ be coherent sheaves on X with $\dim(Supp(\mathcal{F})) \leq r$, $\dim(Supp(\mathcal{G})) \leq s$, and $\dim(Supp(\mathcal{F}) \cap Supp(\mathcal{G})) \leq r + s - \dim X$. In this case $[\mathcal{F}]_r$ and $[\mathcal{G}]_s$ intersect properly and*

$$[\mathcal{F}]_r \cdot [\mathcal{G}]_s = \sum (-1)^p [Tor_p^{\mathcal{O}_X}(\mathcal{F}, \mathcal{G})]_{r+s-\dim(X)}.$$

Proof. The statement that $[\mathcal{F}]_r$ and $[\mathcal{G}]_s$ intersect properly is immediate. Since we are proving an equality of cycles we may work locally on X. (Observe that the formation of the intersection product of cycles, the formation of Tor-sheaves, and

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forming the cycle associated to a coherent sheaf, each commute with restriction to open subschemes.) Thus we may and do assume that X is affine.

Denote

$$RHS(\mathcal{F}, \mathcal{G}) = [\mathcal{F}]_r \cdot [\mathcal{G}]_s \quad \text{and} \quad LHS(\mathcal{F}, \mathcal{G}) = \sum (-1)^p [\mathrm{Tor}_p^{\mathcal{O}_X}(\mathcal{F}, \mathcal{G})]_{r+s-\dim(X)}$$

Consider a short exact sequence

$$0 \to \mathcal{F}_1 \to \mathcal{F}_2 \to \mathcal{F}_3 \to 0$$

of coherent sheaves on X with $\mathrm{Supp}(\mathcal{F}_i) \subset \mathrm{Supp}(\mathcal{F})$, then both $LHS(\mathcal{F}_i, \mathcal{G})$ and $RHS(\mathcal{F}_i, \mathcal{G})$ are defined for $i = 1, 2, 3$ and we have

$$RHS(\mathcal{F}_2, \mathcal{G}) = RHS(\mathcal{F}_1, \mathcal{G}) + RHS(\mathcal{F}_3, \mathcal{G})$$

and similarly for LHS. Namely, the support condition guarantees that everything is defined, the short exact sequence and additivity of lengths gives

$$[\mathcal{F}_2]_r = [\mathcal{F}_1]_r + [\mathcal{F}_3]_r$$

(Chow Homology, Lemma 02QZ) which implies additivity for RHS. The long exact sequence of Tors

$$\ldots \to \mathrm{Tor}_1(\mathcal{F}_3, \mathcal{G}) \to \mathrm{Tor}_0(\mathcal{F}_1, \mathcal{G}) \to \mathrm{Tor}_0(\mathcal{F}_2, \mathcal{G}) \to \mathrm{Tor}_0(\mathcal{F}_3, \mathcal{G}) \to 0$$

and additivity of lengths as before implies additivity for LHS.

By Algebra, Lemma 00L0 and the fact that X is affine, we can find a filtration of $\mathcal{F}$ whose graded pieces are structure sheaves of closed subvarieties of $\mathrm{Supp}(\mathcal{F})$. The additivity shown in the previous paragraph, implies that it suffices to prove $LHS = RHS$ with $\mathcal{F}$ replaced by $\mathcal{O}_V$ where $V \subset \mathrm{Supp}(\mathcal{F})$. By symmetry we can do the same for $\mathcal{G}$. This reduces us to proving that

$$LHS(\mathcal{O}_V, \mathcal{O}_W) = RHS(\mathcal{O}_V, \mathcal{O}_W)$$

where $W \subset \mathrm{Supp}(\mathcal{G})$ is a closed subvariety. If $\dim(V) = r$ and $\dim(W) = s$, then this equality is the **definition** of $V \cdot W$. On the other hand, if $\dim(V) < r$ or $\dim(W) < s$, i.e., $[V]_r = 0$ or $[W]_s = 0$, then we have to prove that $RHS(\mathcal{O}_V, \mathcal{O}_W) = 0$ [1].

Let $Z \subset V \cap W$ be an irreducible component of dimension $r+s-\dim(X)$. This is the maximal dimension of a component and it suffices to show that the coefficient of Z in RHS is zero. Let $\xi \in Z$ be the generic point. Write $A = \mathcal{O}_{X,\xi}$, $B = \mathcal{O}_{X \times X, \Delta(\xi)}$, and $C = \mathcal{O}_{V \times W, \Delta(\xi)}$. By Lemma 0B0T we have

$$\text{coeff of } Z \text{ in } RHS(\mathcal{O}_V, \mathcal{O}_W) = \sum (-1)^i \mathrm{length}_B \mathrm{Tor}_i^B(A, C)$$

Since $\dim(V) < r$ or $\dim(W) < s$ we have $\dim(V \times W) < r + s$ which implies $\dim(C) < \dim(X)$ (small detail omitted). Moreover, the kernel I of $B \to A$ is generated by a regular sequence of length $\dim(X)$ (Lemma 0B0Q). Hence vanishing by Lemma 0B03 because the Hilbert function of C with respect to I has degree $\dim(C) < n$ by Algebra, Proposition 00KQ. □

[1] The reader can see that this is not a triviality by taking $r = s = 1$ and X a nonsingular surface and $V = W$ a closed point x of X. In this case there are 3 nonzero Tors of lengths $1, 2, 1$ at x.